aPrehliadky — User Guide

Photogrammetric 3D capture of bodies and objects with LiDAR · version 2.0 · forensika.eu

1. What the app does

aPrehliadky builds a 3D model of a body or an object from ordinary iPhone frames. With LiDAR, every frame gets depth and gravity, so the model is built directly in meters and can be measured. Everything runs on the phone — no cloud, no account, no upload.

Output: a virtual examination (rotate, zoom, 1 : 1 in space), measurements (mm, cm, dm, m, in, ft), findings anchored to the surface, a PDF report and an .aprehliadka package with hashes as an attachment for the case file or for a record in ePrehliadky (ÚDZS).

> The app is a documentation tool. It supports professional judgement and does not replace it. The user is responsible for the lawfulness of capture and handling of the documentation.

2. What you need

- iPhone or iPad with iOS 18 or later. LiDAR (Pro models) is recommended — it provides scale and depth.
- For computing the model, an A17 Pro / M1 chip or newer (iPhone 15 Pro and later Pro models, iPad M1+). An older device can capture; compute on a supported one (see ch. 9).
- A calibration target printed at 100 % (Menu (…) → Printable calibration target). Check with a ruler: scale 100 mm, QR 40 mm.
- Even diffuse light, no flash. A charged phone — reconstruction takes minutes.

3. New scan

1. Tap +. Enter the case reference, examiner and location (they go into the report).
2. Profile: Deceased — examination (dense capture, accuracy) · Living person (fast capture) · Object / item (stone, tool, trace).
3. Extent: Overview — whole subject, or Detail — region (wound, tattoo). Detail gives a much finer resolution on the same triangle budget.
4. Method: Silent — 4K video (recommended, no shutter sound, depth exactly synchronous) or Photo — full resolution.
5. Place the target flat next to the subject and tap Start capture.

4. Capture

1. Aim at the center of the subject and tap Set scan center. Coverage is computed around this point.
2. Walk around slowly. Frames are captured automatically by camera movement and rotation — nothing to press.
3. Watch the radar in the top right: 36 sectors × 3 heights. White wedge = where you stand; coloured = covered. Green radar = coverage is sufficient.
4. The scan sound rises with coverage, a new sector adds a spark, completion plays a chord — you hear the progress without looking at the screen.
5. Walk around an object at three heights: from below, at level, from above. A body against a wall: accessible sides + from above.
6. Tap Finish capture (min. 20 frames). Processing starts automatically.

> Glossy and wet surfaces are hard for photogrammetry — LiDAR holds the geometry there. Slow down when you see “Moving too fast”.

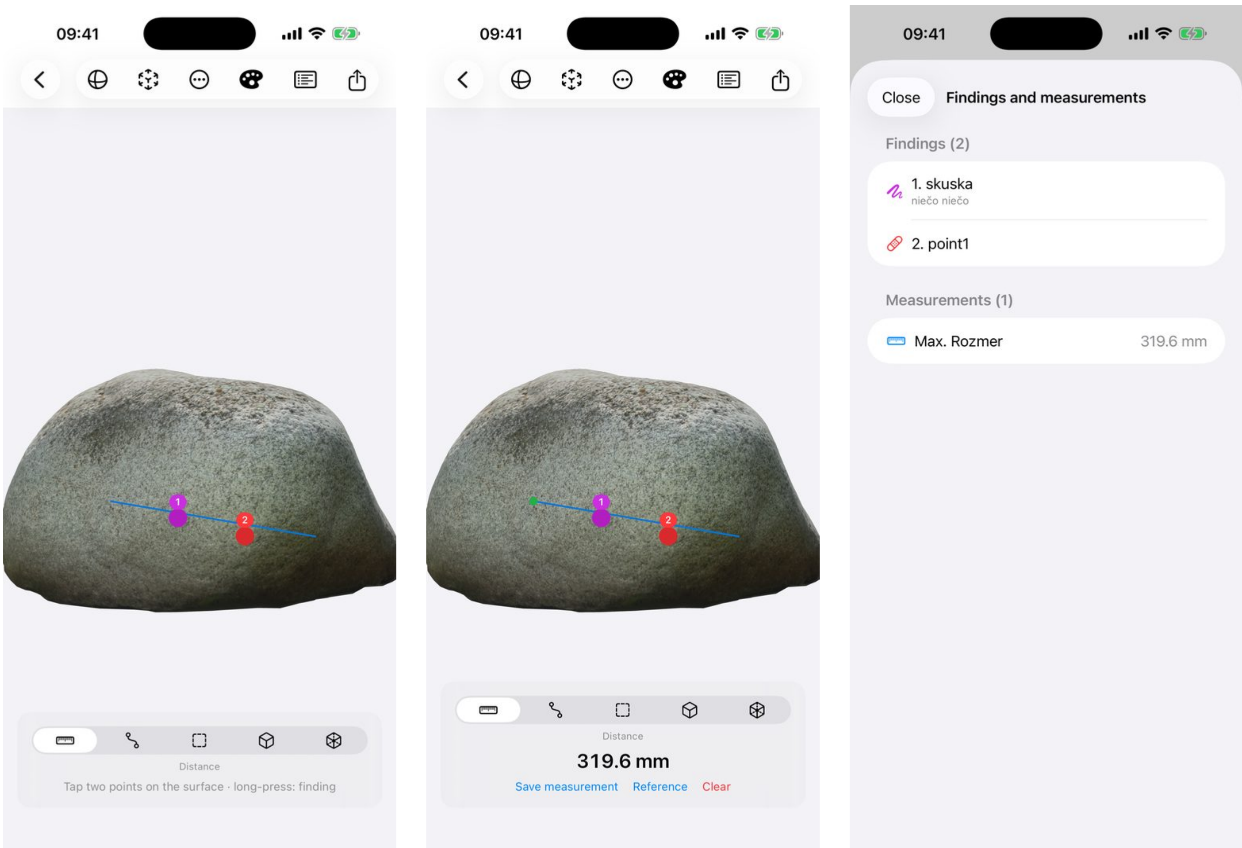
5. Processing

Reconstruction runs entirely on the device and takes minutes. The screen stays on and dimmed (a tap brightens it for 8 s); leave the phone on the charger. A notification arrives when finished and the examination opens.

- Reconstruction extent: Subject (crop to the volume around the center — the whole triangle budget goes to the body) · Subject with surroundings · Whole scene. Cropping runs in two phases; the whole scene remains available as a second examination.
- Sensitivity: High helps with pale skin and low-texture surfaces (takes longer).
- Recompute at any time without a new capture — e.g. a different extent.

6. Virtual examination and measuring

- One finger rotates, two fingers pan, pinch zooms. View menu: front, back, left, right, top, bottom.
- AR 1 : 1 (ARKit icon): place the model on a table at true size and compare with a real ruler.
- Measuring — kind switch: Distance (2 points) · Path (along the surface — circumference, wound length) · Area (polygon ≥ 3 points) · Volume — box (from points) · Volume — model (from the whole mesh, indicative). Tap points on the surface, “Undo point” fixes a slip, “Save measurement” puts it into the report.
- Uneven surfaces: measurement points are anchored to the reconstructed surface (mesh) built from the photographs and LiDAR depth — not to a photograph. Distance is the straight line between two points (chord); if the surface bulges between them, the app also shows the length “along the surface” under the value. Path adds up along the surface: between taps the real mesh point is sampled every ~5 mm, so the length of a wound on a curved body or the circumference of an object follows the actual surface. Where the model has a hole, the straight line is used — a path is never shorter than the chord and nothing is filled in.
- Units: □ → Settings → Measurement units: mm, cm, dm, m, inches (in), feet (ft). The conversion is exact (1 in = 25.4 mm); the package and the report state that values are stored in mm.
- Reference: measure the 100 mm scale of the target, tap Reference and enter 100. The scale deviation of the whole model goes into the report. “Scale vs. LiDAR” (camera trajectory fit, should be ≈ 1.000) gives an independent check.
- Findings: long-press on the surface → type (injury, mark, livor mortis, sample, other), title, description. Numbered labels can be tapped; a finding can be moved (“Move on the model”) or edited.



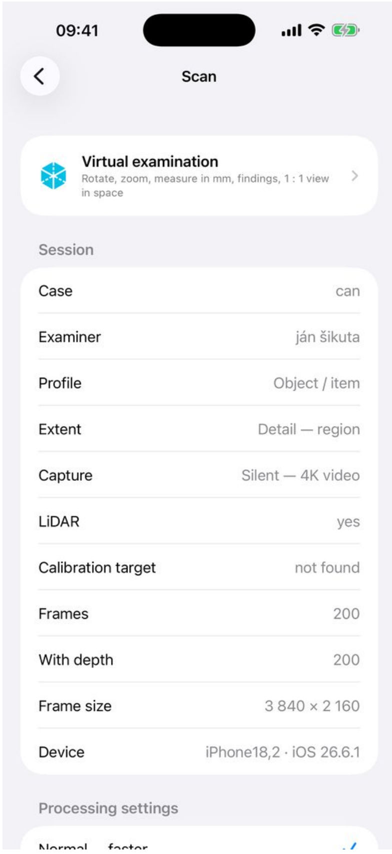
Stone — examination with two findings (labels 1 and 2)

Distance: two points on the surface, 319.6 mm

List of findings and saved measurements

Canister — profile Object / item, extent Detail

Area: four corners of the label → 404.6 cm² (demo)



Scan screen: parameters, processing, diagnostics

Area of a bruise or wound: switch to Area, tap points around the outline of the finding (6–12 points for an irregular shape), watch the value in cm², “Save measurement” with a label (e.g. “bruise left thigh”). The area is computed from the planar projection of the polygon — on a strongly curved surface split it into two smaller ones.

6a. Scale — when a measurement is binding

Photogrammetry alone does not know scale. aPrehliadky obtains it from three independent sources and states in the report which one applies:

1. LiDAR depth (iPhone Pro): every frame carries a depth map embedded in the HEIC, the model is built directly in meters. Report line: “The HEIC carries embedded LiDAR depth”.
2. Camera trajectory (ARKit): camera positions from ARKit are metric even without LiDAR (inertial sensors). After processing they are matched with the photogrammetry poses; the line “Scale vs. LiDAR” should be ≈ 1.000. If the model is not in meters (older sessions, no LiDAR), measurements are automatically scaled by the factor from this fit — shown in the scan as “The model is not in meters — measurements are scaled by a factor of …”.
3. Reference / calibration target: measure the 100 mm scale of the target (or any object of known length), tap Reference and enter the true length. The deviation in % goes into the report. This is the only check a court or a colleague can verify by eye on the model.

With the target: scale verified twice (LiDAR/trajectory + reference) — measurements binding. Without the target, with LiDAR: measurements reliable but without independent proof — state “no reference” in the report. Without the target and without LiDAR: scale only from the camera trajectory (residual in the report); treat measurements as indicative and add a reference afterwards (measure any object of known length in the model).

The examples in this guide (stone, canister) were captured with an older version without embedded depth — the values in the pictures are therefore illustrative; since version 2.0 every HEIC carries depth and the model is in meters.

7. Export, report, archive

- PDF report: case, examiner, capture parameters, scale verification, findings with positions, measurements, integrity, model SHA-256.
- .aprehliadka package: manifest, model, findings, SHA-256 of every file; optionally the frames (allow recomputation elsewhere). The recipient sees on import whether the package was modified.
- Archive: full package to Files → aPrehliadky → Exporty → Archív; the package SHA-256 is written to the manifest and the report. Frames can then be deleted from the phone.
- All exports are in Files → On My iPhone → aPrehliadky → Exporty.

8. ePrehliadky (ÚDZS)

aPrehliadky is designed as a possible add-on to ePrehliadky — the software of the Slovak Health Care Surveillance Authority for electronic post-mortem examinations. The .aprehliadka package and the PDF report are attached to the record; the link aprehliadky://scan/<id> leads back to the scan. The package format is open (docs/FORMAT.md).

aPrehliadky is an independent app by the author. It is not a product of ÚDZS, and ÚDZS neither publishes nor endorses it.

9. Troubleshooting

- The device cannot compute the model: capture, Export → Model, findings and frames, send the package via AirDrop to a supported device, import and start reconstruction.
- I cannot see the model: Scan → Diagnostics → Open in system viewer. If the model exists, it always shows. If not, copy the processing log and send it to jansikuta@me.com.
- Target not found: print at 100 %, target flat, in view of as many frames as possible, no glare.
- The phone heats up: normal during processing; leave it on the charger and at rest.
- Language: → Settings → App language (takes effect after restart).

10. Privacy and donor

The app has no network layer. Frames, models, findings and reports stay on the device until you export them yourself. No analytics, no advertising.

aPrehliadky is free and fully functional with no limits. Anyone who wishes can become a donor (→ Donor) — an optional one-time fee through the App Store that unlocks nothing.